



Advanced Linear Algebra: Foundations to Frontiers

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2.3.4 The Reduced Singular Value Decomposition



02.3.4 Reduced SVD

fit width



Corollary 2.3.4.1. Reduced Singular Value Decomposition. *Let $A \in \mathbb{C}^{m \times n}$ and $r = \text{rank}(A)$. There exist orthonormal matrix $U_L \in \mathbb{C}^{m \times r}$, orthonormal matrix $V_L \in \mathbb{C}^{n \times r}$, and matrix $\Sigma_{TL} \in \mathbb{R}^{r \times r}$ with $\Sigma_{TL} = \text{diag}(\sigma_0, \dots, \sigma_{r-1})$ and $\sigma_0 \geq \sigma_1 \geq \dots \geq \sigma_{r-1} > 0$ such that $A = U_L \Sigma_{TL} V_L^H$.*

Homework 2.3.4.1. Prove the above corollary.

▼ **Solution**

Let $A = U \Sigma V^H = \left(U_L \mid U_R \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(V_L \mid V_R \right)^H$ be the SVD of A , where $U_L \in \mathbb{C}^{m \times r}$, $V_L \in \mathbb{C}^{n \times r}$ and $\Sigma_{TL} \in \mathbb{R}^{r \times r}$ with $\Sigma_{TL} = \text{diag}(\sigma_0, \sigma_1, \dots, \sigma_{r-1})$ and $\sigma_0 \geq \sigma_1 \geq \dots \geq \sigma_{r-1} > 0$. Then

$$\begin{aligned}
A &= \text{< SVD of } A \text{ >} \\
U \Sigma V^T &= \text{< Partitioning >} \\
\left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H \\
&= \text{< partitioned matrix – matrix multiplication >} \\
U_L \Sigma_{TL} V_L^H.
\end{aligned}$$

Corollary 2.3.4.2. Let $A = U_L \Sigma_{TL} V_L^H$ be the Reduced SVD with

$$U_L = \left(\begin{array}{c|c|c} u_0 & \cdots & u_{r-1} \end{array} \right), V_L = \left(\begin{array}{c|c|c} v_0 & \cdots & v_{r-1} \end{array} \right), \text{ and}$$

$$\Sigma_{TL} = \left(\begin{array}{c|c|c} \sigma_0 & & \\ \hline & \ddots & \\ \hline & & \sigma_{r-1} \end{array} \right). \text{ Then}$$

$$A = \sigma_0 u_0 v_0^H + \cdots + \sigma_{r-1} u_{r-1} v_{r-1}^H.$$

Remark 2.3.4.3. This last result establishes that any matrix A with rank r can be written as a linear combination of r outer products:

$$A = \underbrace{\sigma_0 u_0 v_0^H}_{\sigma_0 \mid \text{---}} + \underbrace{\sigma_1 u_1 v_1^H}_{\sigma_1 \mid \text{---}} + \cdots + \underbrace{\sigma_{r-1} u_{r-1} v_{r-1}^H}_{\sigma_{r-1} \mid \text{---}}.$$